Markov Chains and Queueing Systems
EE 621, Spring’26

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Dept. of Electrical Engineering
IIT Bombay

Basic Information

- **Timings:**
  - ❑Monday: 11.35 am to 12.30 pm
  - ❑Tuesday: 8.30 am to 9.25 am
  - ❑Thursday: 9.30 am to 10.25 am
- **Venue:** LH 102
- **Credits:** 6
- **Instructor**: Gaurav S. Kasbekar
  - ❑Office: 211-B, EE Old Building
  - ❑Email: gskasbekar@ee.iitb.ac.in
- **Teaching Assistants**:
  - ❑To be finalized later
- Course material will be posted on Moodle

Prerequisites

- EE 325, EE 601, or equivalent

Course Contents

- Review of Basic Probability Theory, Sequences of Random Variables, Laws of Large Numbers, Central Limit Theorem, Moment Generating Functions and Characteristic Functions
- Renewal Processes, Poisson Process, Elementary Renewal Theorem, Time Averages and Pointwise Limits, Stopping Times and Wald's Equation, Rewards and Costs, Renewal Reward Theorem, Alternating Renewal Processes, The Inspection Paradox, Regenerative Processes, Point Processes, Blackwell's Theorem, Key Renewal Theorem, Applications of Renewal Processes
- Discrete-Time Markov Chains, Transition Probabilities, Chapman-Kolmogorov Equations, Recurrent and Transient States, Communication Classes, Periodic and Aperiodic States, Limit Theorems, Branching Processes, Reversible Markov Chains, Semi-Markov Processes, Stability Analysis Using Lyapunov Functions, Applications of Discrete-Time Markov Chains
- Continuous-Time Markov Chains, Regular Chains, Birth-Death Processes, Kolmogorov's Backward and Forward Equations, Reversibility, Uniformization
- Introduction to Queueing Theory, Little's Theorem, Markovian Queues, Erlang B and C Formulas, Queueing Networks, Tandem Queues, Open and Closed Systems, Jackson Networks, Work, Random Walks, M/G/1, G/M/1, and G/G/1 Queues, Multiserver Queues, Burke's Theorem, Erlang's Loss System, Priority Queues, Vacations, Overflows, Retrials

Textbooks

- S.M. Ross, "*Stochastic Processes*", John Wiley & Sons, Second Edition, 1996.
- R.W. Wolff, "*Stochastic Modeling and the Theory of Queues*", Prentice-Hall, 1989.

Other References

- S.M. Ross, "*Introduction to Probability Models*", Academic Press Inc., 13th Edition, 2023.
- P. Brémaud, "*Markov Chains: Gibbs Fields, Monte Carlo Simulation, and Queues*", Springer, Second Edition, 2020.
- M. Harchol-Balter, "*Performance Modeling and Design of Computer Systems: Queueing Theory in Action",* Cambridge University Press, 2013.
- J.F. Shortle, J.M. Thompson, D. Gross, C.M. Harris, "*Fundamentals of Queueing Theory",* John Wiley & Sons, Fifth Edition, 2018.
- L. Kleinrock, "*Queueing Systems: Vol. 1– Theory*", John Wiley & Sons, 1975.
- L. Kleinrock, "*Queueing Systems: Vol. 2– Computer Applications*", John Wiley & Sons, 1976.
- D. Bertsekas, R. Gallager, "*Data Networks*", Second Edition, Prentice Hall India, 1992.
- Gallager, R.G., *"Stochastic Processes: Theory for Applications",* Cambridge University Press, 2013.
- A. Kumar, "Discrete Event Stochastic Processes", *Lecture Notes for Engineering Curriculum*, 2012.

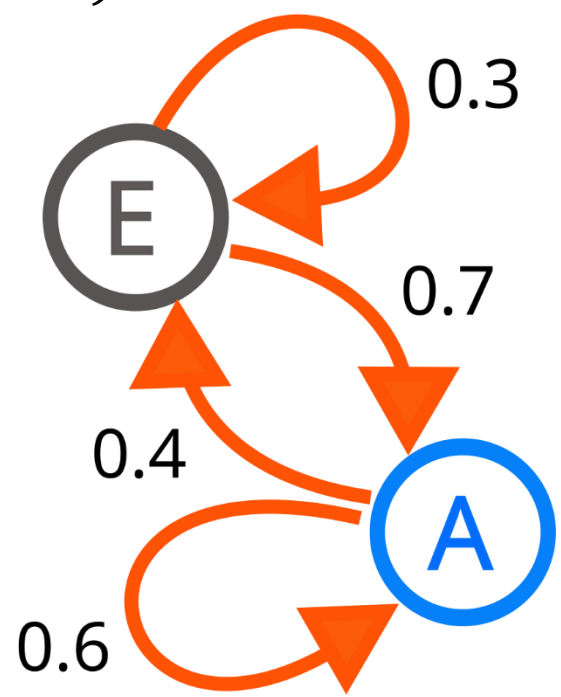
Grading

- Quizzes (best 2 out of 3): 10 % each
- Mid-sem: 30 %
- End-sem: 45 %
- Attendance: 5 %
- Homework (won’t be graded)
  - ❑Teaching assistants will conduct office hours every week for help with homework
  - ❑Can also meet instructor

Motivation

Markov Chains

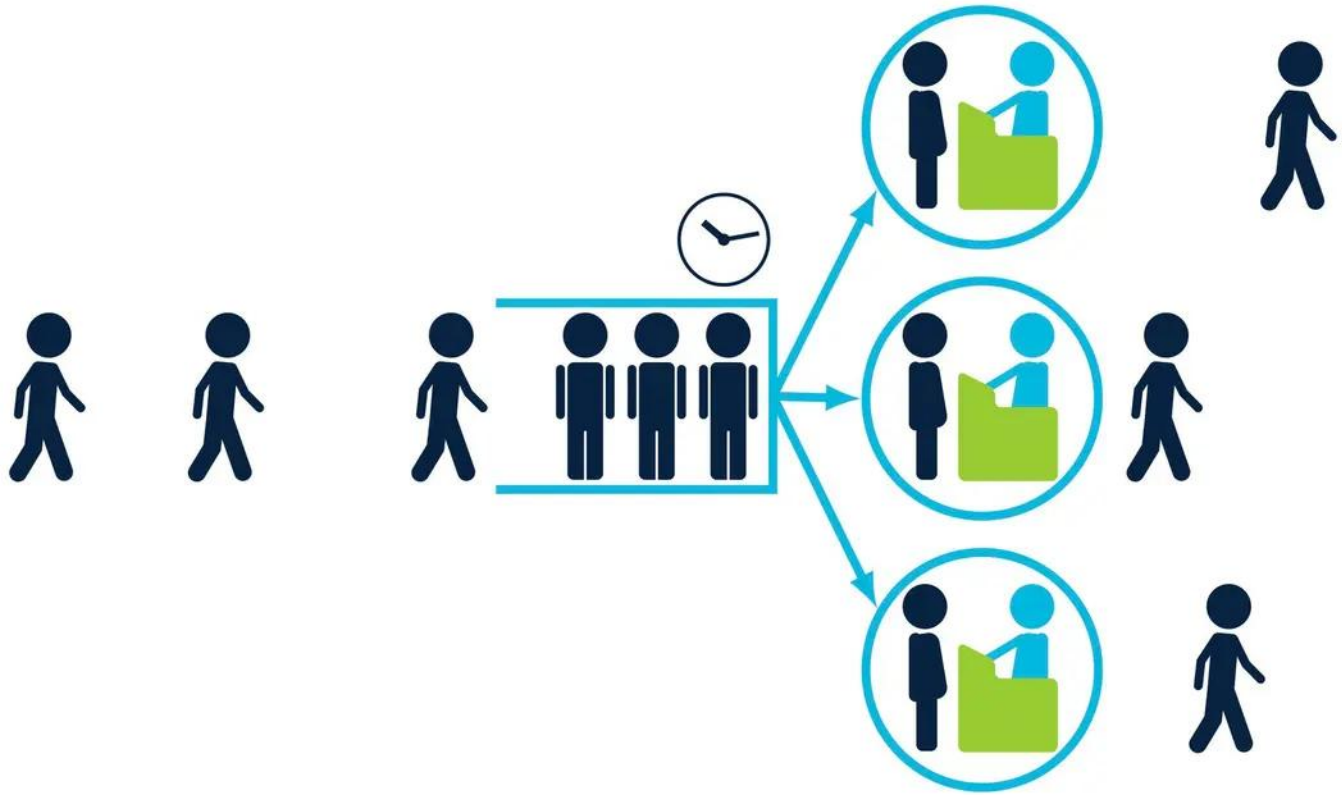
- Consider a stochastic process $\{X_n, n = 0,1,2, \dots\}$
- If $X_n = i$, then process said to be in *state* i at time n
- $\{X_n\}$ said to be a Markov chain if
  - $P(X_{n+1} = j | X_n = i, X_{n-1} = i_{n-1}, \dots, X_1 = i_1, X_0 = i_0) = P(X_{n+1} = j | X_n = i)$ for all states $i_0, i_1, \dots, i_{n-1}, i, j$, and all $n \geq 0$
- E.g.:
  - $P(X_4 = 10 | X_3 = 5, X_2 = 12, X_1 = 3, X_0 = 7) = P(X_4 = 10 | X_3 = 5)$
- *Interpretation*: Given present state, future state is independent of past states



Ref: https://en.wikipedia.org/wiki/Markov_chain

Queueing Theory

- Mathematical theory dealing with analysis of queue behavior

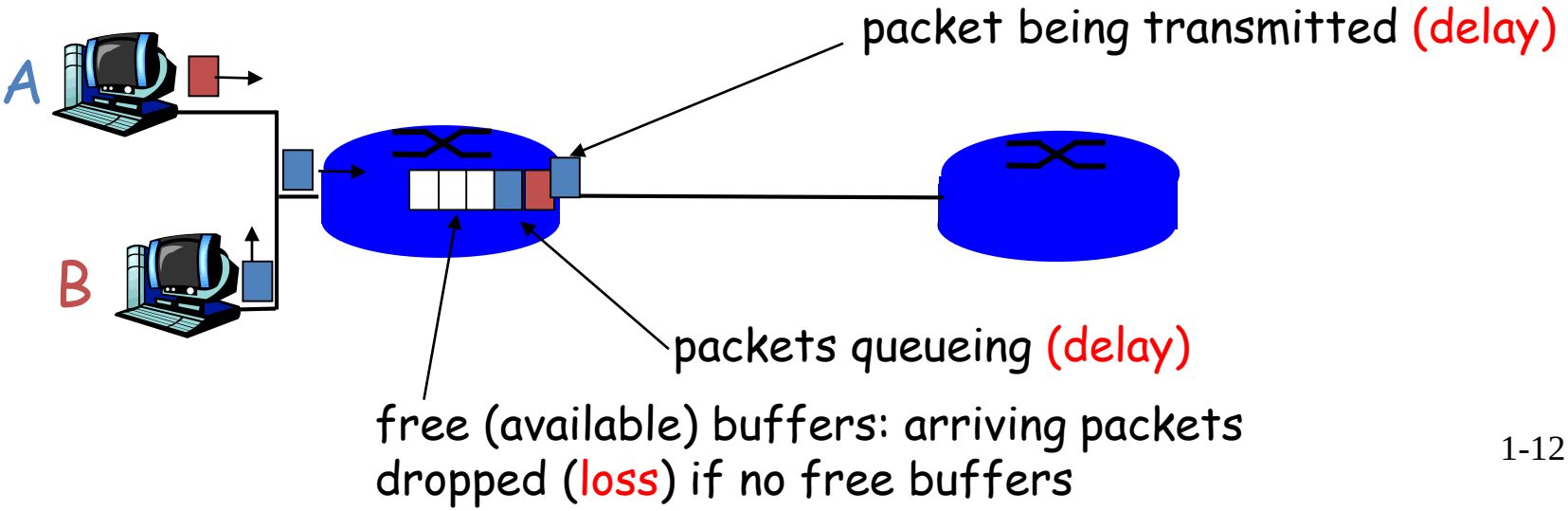


Ref: https://www.edx.org/learn/math/imt-queuing-theory-from-markov-chains-to-multi-server-systems

Applications of Markov Chains and Queueing Theory to Communication Networking

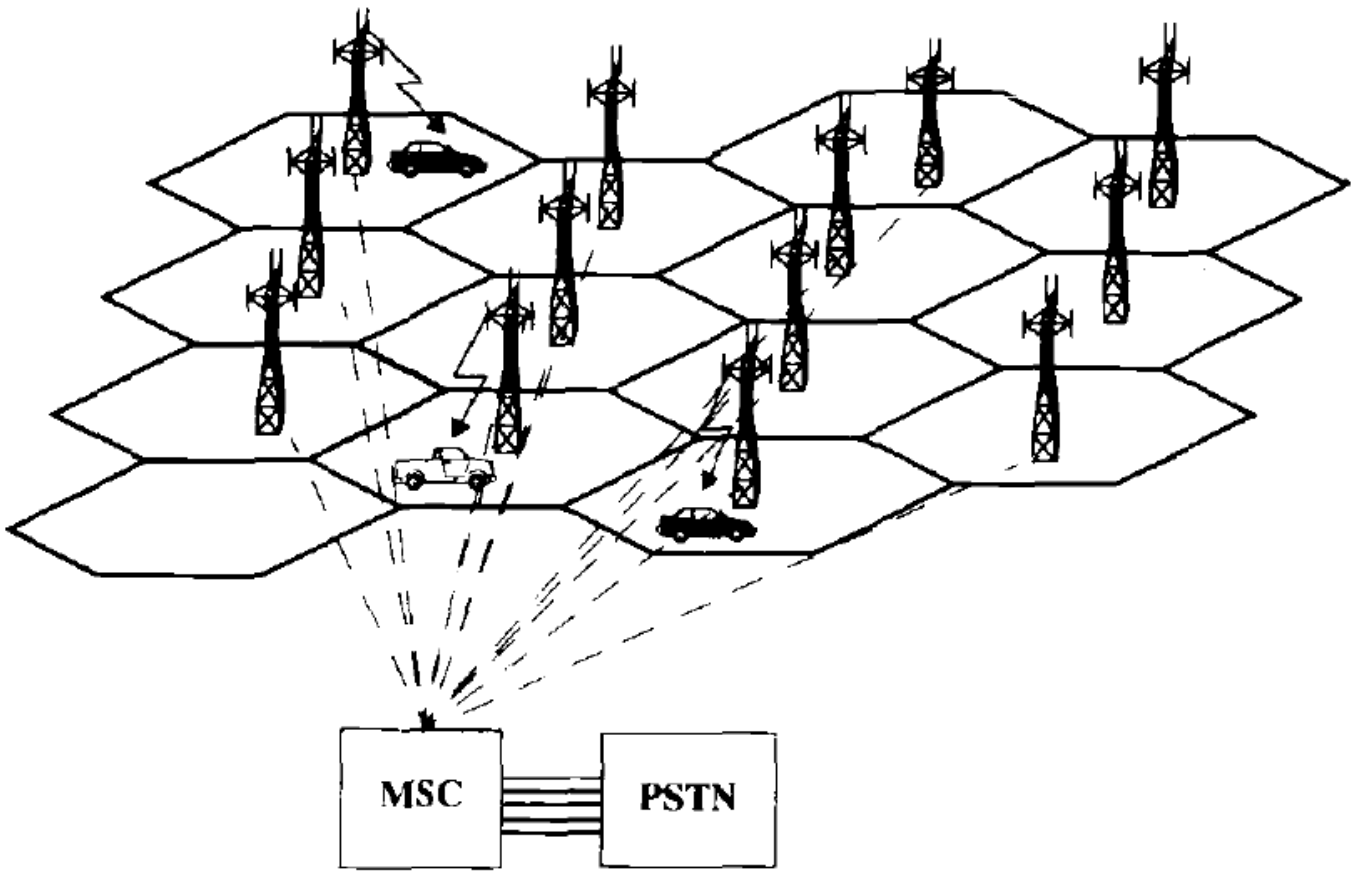
Queueing in Communication Networks

- Packets arrive at router on incoming links
- Router determines output link of each packet from destination IP address
- Transmitted immediately if output link free, else waits in output link buffer for its turn
  - ❑ delay incurred ("*queueing delay*")
- Queueing delays:
  - ❑ can vary significantly from packet to packet
  - ❑ analyzed using queueing theory



Ref: "Computer Networking: A Top Down Approach" by Kurose and Ross

Cellular Network Architecture



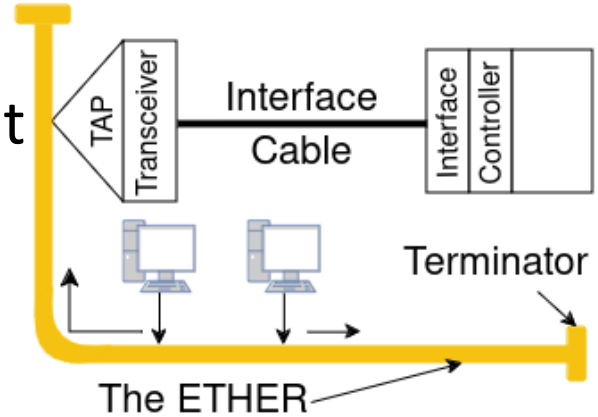
- Region (e.g., city) divided into small "*cells*"
  - typically cell radius < 5 km
- Each cell served by one BS

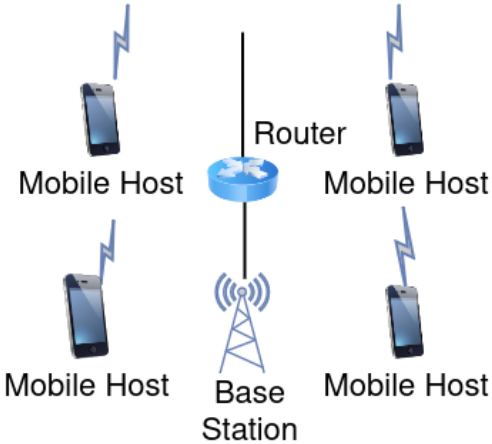
Required Number of Channels Per Cell

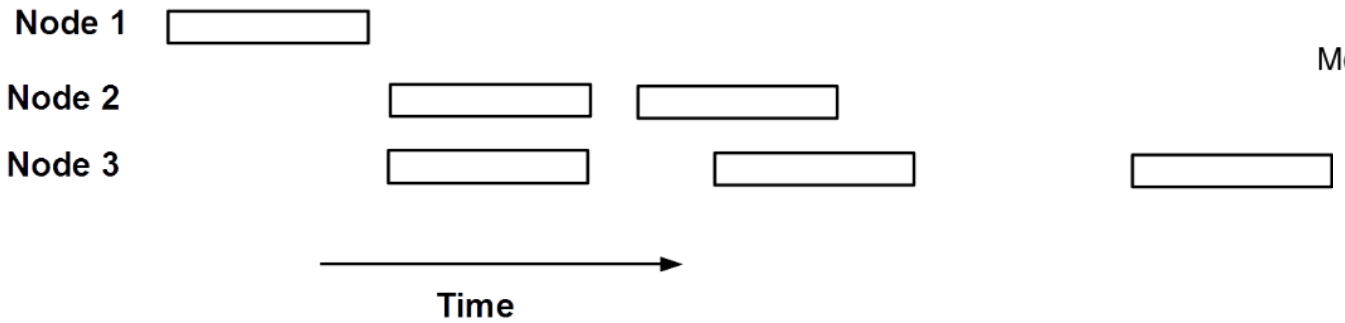
- Consider a fixed cell (e.g., IIT-B campus)
- U subscribers in cell (e.g., 10000)
- k: no. of channels (max. no. of simultaneous calls)
- **Question**: Given U, how much should k be?
- *Not all U subscribers talk simultaneously, so select $k < U$ to save cost*
- If new call arrives when all lines busy, call blocked
- Let P_B be probability that a new call blocked
- *Select k such that P_B less than desired value (e.g., 0.01)*
- Need model to estimate P_B for given population of subscribers
- Based on queueing theory
- Popular model: *Erlang B formula*

Medium Access Control Problem

- N nodes connected to a broadcast medium (e.g., Ethernet, Wi-Fi)
- If exactly one node transmits at a time, packet successfully received
- Collision if transmissions of two or more nodes overlap
- Need a way to make sure that exactly one node transmits at a time
- Medium Access Control (MAC) Protocol:
  - ❑ distributed algorithm that determines how nodes share channel, i.e., determines when given node can transmit

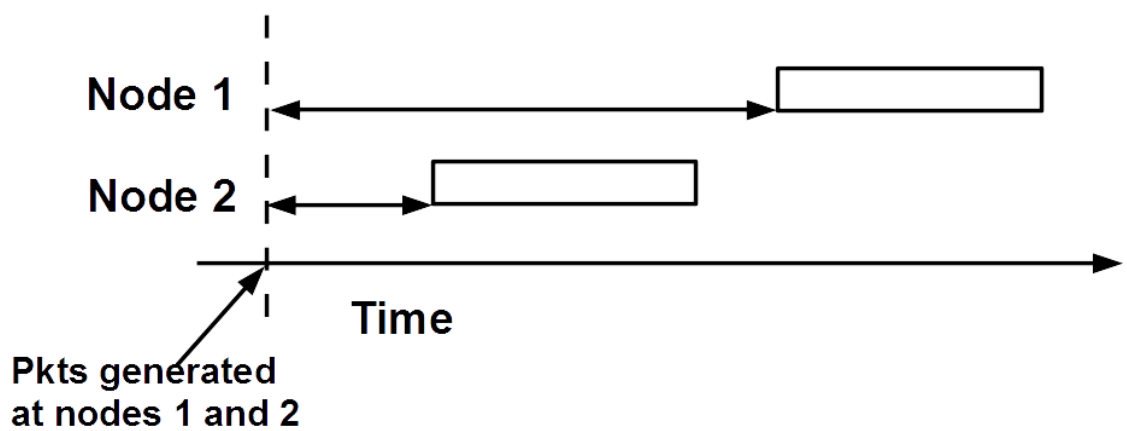






Randomized MAC Protocols

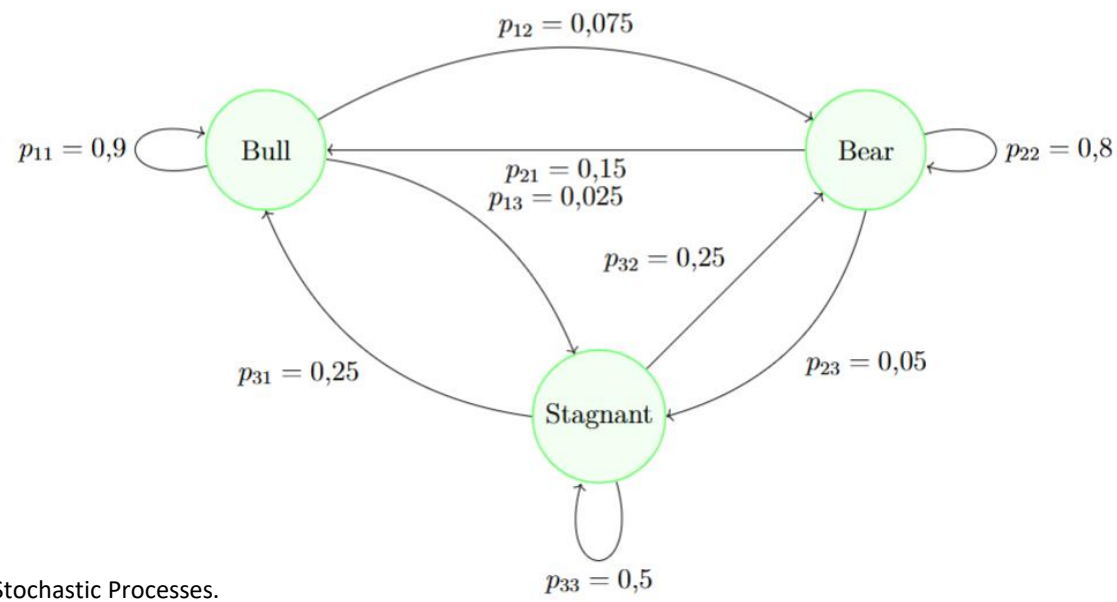
- Suppose packets generated at nodes 1 and 2 at roughly same time
- If they transmit immediately, collision
  - ❑ if they retransmit, repeated collisions
- If each waits for a random amount of time before transmitting, successful transmission with some probability
- E.g. of randomized MAC protocols:
  - ❑ Pure ALOHA and Slotted ALOHA (satellite networks),
  - ❑ CSMA/ CA (Wi-Fi),
  - ❑ CDMA/ CD (Ethernet)
- Markov chains and queueing theory used for analysis, e.g.:
  - ❑ Markov chain model for:
    - remaining wait time of each node, number of consecutive collisions that have occurred
  - ❑ queue of packets to be sent at each node



Some More Applications of Markov Chains

Finance and Economics

- Stock Market Regimes:
  - ❑Investors use Markov chains to model different market "states“:
    - ○ e.g., "Bull," "Bear," or "Stagnant"
- Credit Ratings:
  - ❑Banks use Markov chains to model different categories of borrower's credit score:
    - ○e.g., AAA, AA, or Default
  - ❑This helps them set interest rates



Ref: Fedotova, A., 2021. Data Assimilation for Discrete-time Stochastic Processes.

Biology and Genetics

- Gene Expression:
  - how cells turn instructions in DNA into functional products, like proteins, that fight disease, build tissues, etc.
  - Markov chains used to model gene expression by treating:
    - biological stages (DNA, mRNA, Protein) as states and
    - transitions between them (transcription, translation) as probabilistic steps
- Population Genetics:
  - Wright-Fisher Model uses Markov chains to describe how gene frequencies change in a population over generations

Natural Language Processing (NLP)

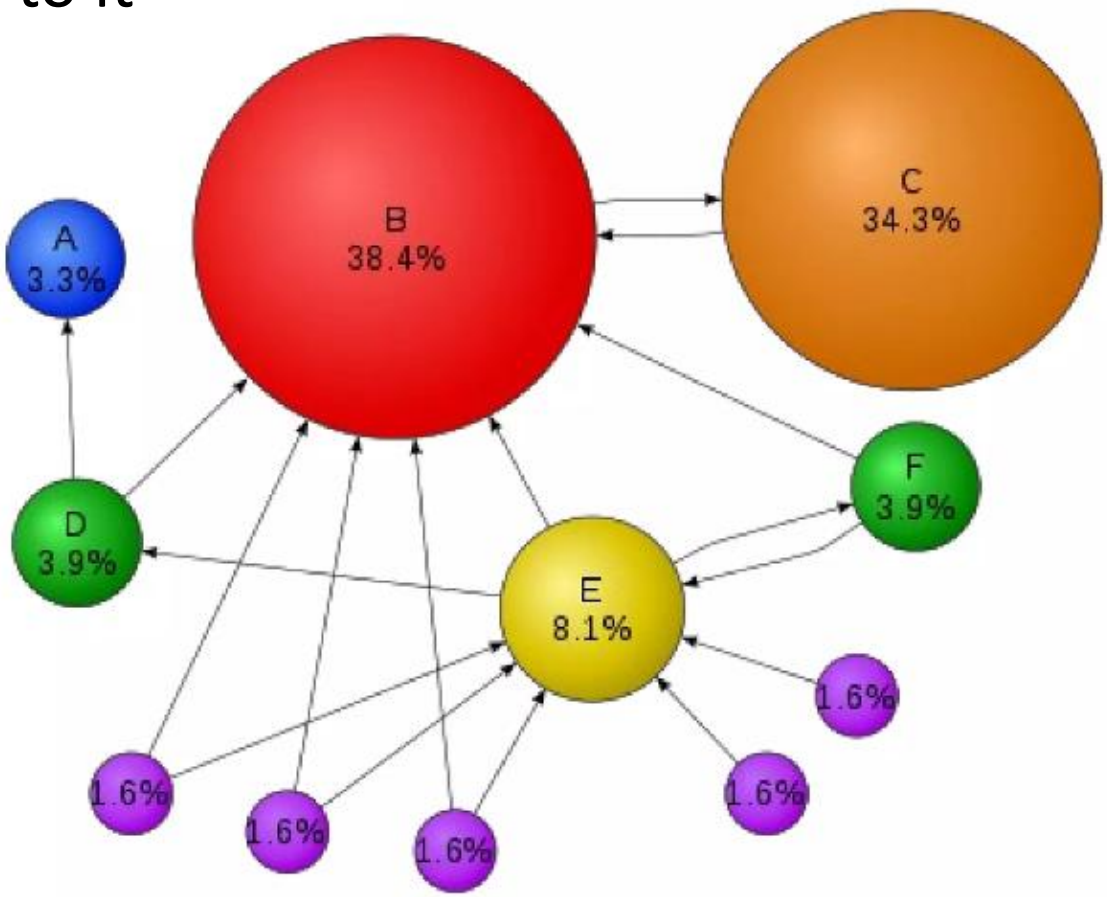
- Predictive Text:
  - When your phone suggests next word you might type, it is often using a Markov model
  - It looks at current word ("How") and calculates probability of next state ("are," "is," or "was")
- Speech Recognition:
  - Hidden Markov Models (HMMs) used to translate acoustic signals (sounds) into sequence of words
  - Sounds treated as "observations" of hidden linguistic states

Meteorology and Weather Forecasting

- A simple Markov model might have two states: "Rain" and "No Rain"
- This model can be expanded to include "Clear," "Cloudy”, "Foggy”, and "Stormy" to create more complex weekly forecasts

Google PageRank Algorithm

- PageRank treats entire web as state space where each page is a state
- User clicks a link on a page to move to a new one
- Probability of landing on specific page is determined by number of links pointing to it
- A page is considered important if many other important pages transition to it
- By calculating "steady state" of this massive Markov chain, Google can rank pages by their importance



Ref: https://www.slideshare.net/slideshow/markov-chain-and-its-application/107079861

Some More Applications of Queueing Theory

Computing

- Server Performance:
  - queue of user requests at web server
  - web servers need to determine how many simultaneous users they can handle before site crashes or slows down
- Operating Systems:
  - queue of tasks at CPU
  - need to decide which task should be served by CPU next to minimize wait times

Healthcare Management

- Emergency Rooms:
  - patients are not served first-come-first-served, but based on severity of their condition (priority queueing)
- Resource Planning:
  - administrators need to decide how many beds, nurses, or MRI machines are needed to keep patient wait times within safe limits
- Organ Transplants:
  - matching process for donors and recipients modeled as complex queue

Transportation and Urban Planning

- Traffic Lights:
  - ❑ queueing models used to set timing of green and red lights
- Airport Runways:
  - ❑ air traffic controllers treat runways as servers and airplanes as customers
  - ❑ queueing theory helps schedule takeoffs and landings to minimize fuel waste and delays
- Toll Booths and Parking:
  - ❑ determining number of toll lanes
  - ❑ determining size of parking garage

Ref: https://www.engineersgarage.com/wp-content/uploads/2025/04/TCH174-01-How-Smart-Traffic-Light-System-Works.jpeg

Ref: https://www.financialexpress.com/business/roadways-toll-free-pass-for-vehicles-in-longer-than-100-m-queue-2259782/

Manufacturing and Logistics

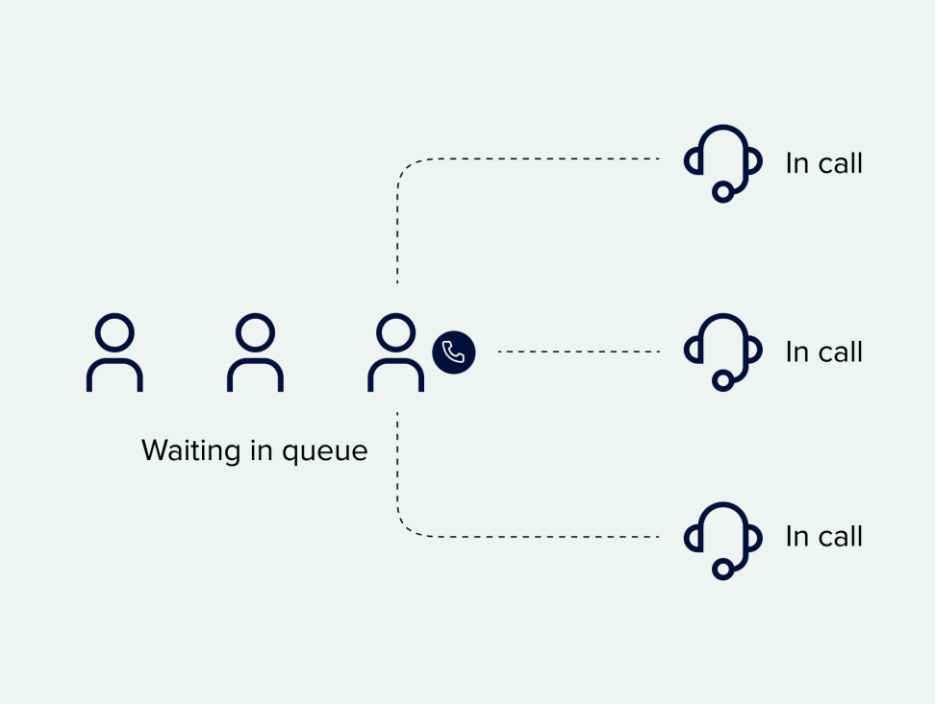
- Assembly Lines in Factory:
  - queues of parts to be processed waiting at different machines
- Warehousing:
  - networks of "servers" (loading docks, forklifts, pickers) and "customers" (incoming trucks, orders)
- Just-in-Time (JIT) Inventory:
  - to keep just enough stock to meet arrivals without paying for excess storage

Ref: https://mikhail.io/2018/11/from-0-to-1000-instances-how-serverless-providers-scale-queue-processing/

Retail and Service Industry

- Bank and Grocery Lines:
  - ❑ customers in queues for banking services or checkout
- Call Centers:
  - ❑ queue of calls at agents
  - ❑ queueing theory determines number of agents needed during peak hours
  - ❑ used to generate "estimated wait time" messages heard when on hold
- Theme Parks (e.g., Disneyland):
  - ❑ queues of visitors at different rides

Ref: https://stock.adobe.com/



Ref: https://dialerhq.com/blog/call-queuing/